Dr Sanjay Modgil

sanjay.modgil@kcl.ac.uk

https://sanjaymodgil.nms.kcl.ac.uk/

Informatics

Natural, Mathematical & Engineering Sciences

KING'S College LONDON

Non-monotonic Reasoning, Argumentation and Dialogue

Outline

- Week 2

- Three example non-monotonic logics – *syntactic* rather than *semantic* presentations

- Formalising non-monotonic reasoning in terms of the exchange of *arguments*

  - Argumentation Frameworks: Constructing arguments and counter-arguments from a set of formulae
  - Using preferences to decide when one argument legitimately challenges another
  - Labelling rules to decide which are the winning arguments

Outline

- Week 3

  - *Argument Game Proof theories* to decide whether an argument is winning

  - From argument game proof theories to *dialogues for* ***distributed*** *non-monotonic reasoning*: Multiple agents communicating so as to pool their knowledge and reason together

  - The importance of dialogue in the dawning age of Artificial Intelligence

Week 2 Learning Outcomes

By the end of this week, you will be able to:

- LO1: Evaluate the non-monotonic consequences for three types of non-monotonic logic
- LO2: Construct an argument framework (AF) of arguments and their relations from a knowledge base (KB) of formulas and the inference rules of a logic
- LO3: Apply preferences to determine whether one argument is a legitimate counter-argument to another
- LO4: Apply labelling rules to determine the winning sets of arguments in an AF
- LO5: Relate the claims of winning arguments to KB's non-monotonic logic consequences

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- LO5: Relate the claims of winning arguments to KB's non-monotonic logic consequences

Logic Programming

- You have studied logic programming and Prolog in 4CCSA1LKR
- Recall that logic programs do not have strict negation ($\neg$) but instead negation as failure (*NOT*)

KB_1

1. *penguin(tweety)*
2. *bird(X) :- penguin(X)*
3. *fly(X) :- bird(X), NOT abnormal_fly(X)*

$\mid\!\sim$ *fly(tweety)*

- Intuitively, in the most preferred/normal worlds (with respect to the property of flying) in which X is a bird, X flies

Logic Programming

$\mathbf{KB_2 = KB_1} \cup \{4\}$

1. *penguin(tweety)*
2. *bird(X) :- penguin(X)*
3. *fly(X) :- bird(X), NOT abnormal_fly(X)*
4. *abnormal_fly(X) :- penguin(X), NOT DC_comic(X)*

$\not\models$ *fly(tweety)*

- The most normal/preferred worlds in which X is a penguin (and not the DC character) are not the most normal worlds with respect to the property of flight

Logic Programming

$\mathbf{KB_3 = KB_2} \cup \{5\}$

1. *penguin(tweety)*
2. *bird(X) :- penguin(X)*
3. *fly(X) :- bird(X), NOT abnormal_fly(X)*
4. *abnormal_fly(X) :- penguin(X), NOT DC_comic(X)*
5. *DC_comic(tweety)*

$\mid\!\sim$ *fly(tweety)*

Maxiconsistent Approaches to Non-monotonic Reasoning

- Recall that classical logic (propositional or predicate) is **monotonic**

$$\mathbf{KB} \vdash \alpha \quad \longrightarrow \quad \mathbf{KB} \cup \{\beta\} \vdash \alpha$$

Maxiconsistent Approaches to Non-monotonic Reasoning

- Recall that classical logic (propositional or predicate) is **monotonic**

$$\mathbf{KB} \vdash \alpha \quad \longrightarrow \quad \mathbf{KB} \cup \{\beta\} \vdash \alpha$$

- But one can also think of non-monotonic reasoning as a decision procedure for choosing between contradictory **classical** (monotonic) consequences

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- Recall that classical logic (propositional or predicate) is **monotonic**

$$\mathbf{KB} \vdash \alpha \;\longrightarrow\; \mathbf{KB} \cup \{\beta\} \vdash \alpha$$

- But one can also think of non-monotonic reasoning as a decision procedure for choosing between contradictory **classical** (monotonic) consequences

KB_1

1. *penguin(tweety)*
2. *penguin(X)* $\longrightarrow$ *bird(X)*
3. *bird(X)* $\longrightarrow$ *fly(X)*

$\vdash$ *fly(tweety)*

KB_2

1. *penguin(tweety)*
2. *penguin(X)* $\longrightarrow$ *bird(X)*
3. *bird(X)* $\longrightarrow$ *fly(X)*
4. *penguin(X)* $\longrightarrow \neg$*fly(X)*

$\vdash$ *fly(tweety)*

$\vdash$ $\neg$fly(tweety)

Maxiconsistent Approaches to Non-monotonic Reasoning

KB_1

1. *penguin(tweety)*
2. *penguin(X)* $\longrightarrow$ *bird(X)*
3. *bird(X)* $\longrightarrow$ *fly(X)*

$\vdash$ *fly(tweety)*

KB_2

1. *penguin(tweety)*
2. *penguin(X)* $\longrightarrow$ *bird(X)*
3. *bird(X)* $\longrightarrow$ *fly(X)*
4. *penguin(X)* $\longrightarrow$ $\neg$*fly(X)*

$\vdash$ *fly(tweety)*

$\vdash$ $\neg$fly(tweety)

- Intuitively, since penguins are a special type of (subclass of) of bird (*penguin(X)* $\longrightarrow$ *bird(X)*) they may not inherit all the properties of birds. They are exceptional (`abnormal') with respect to the property of flying
- This 'subclass principle' justifies the consequence $\neg$*fly(tweety)* in preference to *fly(tweety)* by prioritising the implication 4 over the implication 3
- This suggests that we can obtain a non-monotonic consequence relation by prioritising certain formulas over other formulas and then, when consequences contradict, choosing those consequences that make use of formulas with higher priority

Maxiconsistent Approaches to Non-monotonic Reasoning

KB_1

1. *penguin(tweety)*
2. *penguin(X) ⟶ bird(X)*
3. *bird(X) ⟶ fly(X)*

⊢ *fly(tweety)*

KB_2

1. *penguin(tweety)*
2. *penguin(X) ⟶ bird(X)*
3. *bird(X) ⟶ fly(X)*
4. *penguin(X) ⟶ ¬fly(X)*

⊢ *fly(tweety)*

⊢ ¬fly(tweety)

- We can assume a total ordering ≥ over classical logic formulas in a KB. Recall that:

$$\boldsymbol{x \geq y \text{ and } y \not\geq x \rightarrow x > y}$$
$$\boldsymbol{x \geq y \text{ and } y \geq x \rightarrow x \approx y}$$

- Facts and formulas expressing subclass relationships have highest priority and belong to same equivalence class. Formulas expressing properties have less priority and are ordered relative to each other based on subclass relationship

Maxiconsistent Approaches to Non-monotonic Reasoning

1. *penguin(tweety)*
2. *penguin(X)* $\longrightarrow$ *bird(X)*
3. *bird(X)* $\longrightarrow$ *fly(X)*
4. *penguin(X)* $\longrightarrow \neg$*fly(X)*

$1 \geq 2$ and $2 \geq 1$ *(and so* $1 \approx 2$ *)*

$1, 2 \geq 3, 4$ *(and so* $1 > 3,4$ *and* $2 > 3,4$ *)*

$4 \geq 3$ *(and so* $4 > 3$ *)*

- We can represent totally ordered formulas as a stratified KB with each stratum representing an equivalence class and formulas in a higher stratum strictly greater than those in a lower stratum (there are other criteria for ordering formulas, e.g., temporal ordering of formulas)

S_1	*penguin(tweety)* *penguin(X)* $\longrightarrow$ *bird(X)*
S_2	*penguin(X)* $\longrightarrow \neg$*fly(X)*
S_3	*bird(X)* $\longrightarrow$ *fly(X)*

Maxiconsistent Approaches to Non-monotonic Reasoning

S_1	*penguin(tweety)* *penguin(X)* $\longrightarrow$ *bird(X)*
S_2	*penguin(X)* $\longrightarrow \neg$*fly(X)*
S_3	*bird(X)* $\longrightarrow$ *fly(X)*

- Given a stratified KB $S_1 \ldots S_n$ select the maximal consistent subsets in a way that respects the prioritisation of formulas
- Intuitively, identify the maximal consistent subsets $E_1 \ldots E_m$ of S_1 and for each E_i maximally consistently extend with formulas from S_2 and so on …

Maxiconsistent Approaches to Non-monotonic Reasoning

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- Intuitively, identify the maximal consistent subsets $E_1 \ldots E_m$ of S_1 and for each E_i maximally consistently extend with formulas from S_2 and so on …

- The procedure is best explained by example …

S_1	$a \quad a \longrightarrow b \quad \neg b$
S_2	$c \quad \neg c$
S_3	d

$E_1 = \{a\ ,\ a \longrightarrow b\}$ $\quad E_2 = \{a\ ,\ \neg b\ \}$ $\quad E_3 = \{a \longrightarrow b\ ,\ \neg b\ \}$

Maxiconsistent Approaches to Non-monotonic Reasoning

- Given a stratified KB $S_1 \ldots S_n$ select the maximal consistent subsets in a way that respects the prioritisation of formulas

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S_1	$a \quad a \longrightarrow b \quad \neg b$
S_2	$c \quad \neg c$
S_3	d

$E_1 = \{a\ ,\ a \longrightarrow b\}$

$E_{1.1} = \{a\ ,\ a \longrightarrow b,\ c\}$
$E_{1.2} = \{a\ ,\ a \longrightarrow b, \neg c\}$

$E_{1.1.3} = \{a\ ,\ a \longrightarrow b,\ c,\ d\}$
$E_{1.2.3} = \{a\ ,\ a \longrightarrow b, \neg c,\ d\}$

$E_2 = \{a\ ,\ \neg b\ \}$

$E_{2.1} = \{a\ ,\ \neg b,\ c\}$
$E_{2.2} = \{a\ ,\ \neg b, \neg c\}$

$E_{2,1,3} = \{a\ ,\ \neg b,\ c,\ d\}$
$E_{2.2,3} = \{a\ ,\ \neg b,\ \neg c,\ d\}$

$E_3 = \{a \longrightarrow b\ ,\ \neg b\ \}$

$E_{3.1} = \{a \longrightarrow b,\ \neg b,\ c\}$
$E_{3.2} = \{a \longrightarrow b,\ \neg b,\ \neg c\}$

$E_{3.1,3} = \{a \longrightarrow b,\ \neg b,\ c,\ d\}$
$E_{3.2,3} = \{a \longrightarrow b,\ \neg b,\ \neg c,\ d\}$

Maxiconsistent Approaches to Non-monotonic Reasoning

S_1	a	$a \longrightarrow b$	$\neg b$
S_2	c	$\neg c$	
S_3	d		

$E_1 = \{a\ ,\ a \longrightarrow b\}$

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- Let $\geq$ be a total ordering over KB and $S_1 \ldots S_n$ the stratification of KB such that

  - $\alpha, \beta \in S_i$ *iff* $\alpha \approx \beta$
  - $\alpha \in S_i$, $\beta \in S_j$ and $i < j$ iff $\alpha > \beta$

- a *preferred maximal consistent subset* (*mcs*) is a set $E = E_1 \cup \ldots \cup E_n$ such that:
  for $i = 1, \ldots, n$, $E_1 \cup \ldots \cup E_i$ is a maximal (under set inclusion) consistent subset of $S_1 \ldots S_n$

Maxiconsistent Approaches to Non-monotonic Reasoning

S_1	$a \quad a \longrightarrow b \quad \neg b$
S_2	$c \quad \neg c$
S_3	d

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$E_{1.1} = \{a\ ,\ a \longrightarrow b,\ c\}$
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Maxiconsistent Approaches to Non-monotonic Reasoning

KB_1

s_1	*penguin(tweety)*	*penguin(X)* $\longrightarrow$ *bird(X)*
s_2	*bird(X)* $\longrightarrow$ *fly(X)*	

- There is a single preferred mcs: E = { *penguin(tweety)* , *penguin(X)* $\longrightarrow$ *bird(X)* , *bird(X)* $\longrightarrow$ *fly(X)* }

$$KB_1 \mid\!\sim \textit{fly(tweety)}$$

Maxiconsistent Approaches to Non-monotonic Reasoning

$KB_2 = KB_1 \cup$
$\{penguin(X) \longrightarrow \neg fly(X)\}$

S_1	$penguin(tweety) \quad penguin(X) \longrightarrow bird(X)$
S_2	$penguin(X) \longrightarrow \neg fly(X)$
S_3	$bird(X) \longrightarrow fly(X)$

- There is a single preferred mcs: E = { $penguin(tweety)$, $penguin(X) \longrightarrow bird(X)$, $penguin(X) \longrightarrow \neg\, fly(X)$ }

$$KB_2 \not\mid\!\sim \; fly(tweety)$$

$$KB_2 \mid\!\sim \; \neg\, fly(tweety)$$

Maxiconsistent Approaches to Non-monotonic Reasoning

- What if there are more than one preferred mcs ?

KB

S_1	$a \quad a \longrightarrow b \quad \neg b$
S_2	$c \quad \neg c$
S_3	d

mcs(KB) =

$E_{1.1.3} = \{a\ ,\ a \longrightarrow b,\ c,\ d\}$
$E_{1.2.3} = \{a\ ,\ a \longrightarrow b, \neg c,\ d\}$
$E_{2,1,3} = \{a\ ,\ \neg b,\ c,\ d\}$
$E_{2.2,3} = \{a\ ,\ \neg b,\ \neg c,\ d\}$
$E_{3.1,3} = \{a \longrightarrow b,\ \neg b,\ c,\ d\}$
$E_{3.2,3} = \{a \longrightarrow b,\ \neg b,\ \neg c,\ d\}$

- We can either define a *credulous* consequence relation:

$$\text{KB} \mid\!\sim_{cred} \alpha \quad \text{iff} \quad \exists\, E \in mcs(KB) \text{ such that } E \vdash \alpha$$

Eg KB $\mid\!\sim_{cred} a, \neg a, b, \neg b, c, \neg c \ \ldots.$ KB $\not\mid\!\sim_{cred} \bot$

Maxiconsistent Approaches to Non-monotonic Reasoning

- What if there are more than one preferred mcs ?

KB

S_1	$a \quad a \longrightarrow b \quad \neg b$
S_2	$c \quad \neg c$
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$E_{1.1.3} = \{a\ ,\ a \longrightarrow b,\ c,\ d\}$
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- Or a *sceptical* consequence relation (this is the consequence relation assumed by Kraus, Lehman Magidor – recall Chris in Lecture 1)

$$\text{KB} \mid\!\sim_{scept} \alpha \quad \text{iff} \quad \forall\, E \in mcs(KB) : E \vdash \alpha$$

Eg $\text{KB} \mid\!\sim_{scept} d \qquad \text{KB} \not\mid\!\sim_{scept} a, \neg a, b, \neg b, c, \neg c$

Maxiconsistent Approaches to Non-monotonic Reasoning

KB_1

s_1	*penguin(tweety)*	*penguin(X)* $\longrightarrow$ *bird(X)*
s_2	*bird(X)* $\longrightarrow$ *fly(X)*	

$$KB_1 \mid\!\sim_{scept\,/\,cred} fly(tweety)$$

$KB_2 = KB_1 \cup$
$\{penguin(X) \longrightarrow \neg fly(X)\}$

s_1	*penguin(tweety)*	*penguin(X)* $\longrightarrow$ *bird(X)*
s_2	*penguin(X)* $\longrightarrow \neg$ *fly(X)*	
s_3	*bird(X)* $\longrightarrow$ *fly(X)*	

$$KB_2 \not\mid\!\sim_{scept\,/\,cred} fly(tweety) \qquad KB_2 \mid\!\sim_{scept\,/\,cred} \neg\, fly(tweety)$$

Default Logic Approaches to Non-monotonic Reasoning

- The basic idea is that you have a ***consistent*** factual description of the world Cn(*W*) where W is set of classical formulas …

W =

$$penguin(tweety)$$
$$penguin(X) \rightarrow bird(X)$$

- … and a set of default rules of the form $d = \alpha_1, \ldots, \alpha_n \Longrightarrow \beta$

  where *antecedents*(*d*) = $\alpha_1, \ldots, \alpha_n$ and *consequent*(*d*) = β

- which express that if $\alpha_1, \ldots, \alpha_n$ then usually/typically/under normal circumstances it follows that β

$$bird(X) \Rightarrow fly(X)$$
$$penguin(X) \Longrightarrow \neg fly(X)$$
$$\top \Longrightarrow wings(X)$$

Default Logic Approaches to Non-monotonic Reasoning

- (W, *Defaults* = $\{d_1,\ldots,d_n\}$)

- A default extension E of (W, *Defaults)* is defined as:

$$1.\, E = Cn\left(W \cup \bigcup_{d_i \in Defaults} consequent(d_i)\ s.t.\, antecedents(d) \subseteq E\right)\ and$$

$$2.\, E \nvdash \bot\ and$$

$$3.\ E\ is\ a\ maximal\ (under\ set\ inclusion)\ set\ satisfying\ 1\ and\ 2$$

- Intuitively choose some d_i whose antecedents are in Cn(W) and add consequent β of d_i if one can do so consistently, obtaining Cn($W \cup \{\beta\}$), then choose some d_j whose antecedents are in Cn($W \cup \{\beta\}$) and add consequent δ of d_j if one can do so consistently, obtaining Cn($W \cup \{\beta, \delta\}$), and so on ….

Default Logic Approaches to Non-monotonic Reasoning

KB_1

W =

$$penguin(tweety)$$
$$penguin(X) \rightarrow bird(X)$$

Defaults =

$$bird(X) \Rightarrow fly(X)$$
$$penguin(X) \Longrightarrow \neg fly(X)$$
$$\top \Longrightarrow wings(X)$$

$$E_1 = Cn(W \cup \{wings(tweety), fly(tweety)\})$$

$$E_2 = Cn(W \cup \{wings(tweety), \neg fly(tweety)\})$$

$KB_1 \not\mid\!\sim_{scept}$ *fly(tweety)* , $\neg$ *fly(tweety)* $KB_2 \mid\!\sim_{cred}$ *fly(tweety)* , $\neg$ *fly(tweety)*

Default Logic Approaches to Non-monotonic Reasoning

KB_2

W =

$penguin(X)$
$penguin(X) \rightarrow bird(X)$

Defaults +
Linear
Priority Ordering =

$1. bird(X) \Rightarrow fly(X)$
$2. penguin(X) \Longrightarrow \neg fly(X)$
$\top \quad \Longrightarrow wings(X)$
2 > 1

When choosing defaults start with highest prioritised default whose antecedents hold, then add consequent, then choose next highest prioritised default whose antecedents hold, and so on ….

$$E_2 = Cn(W \cup \{wings(X), \neg fly(X)\})$$

$KB_2 \not\mid\!\sim_{scept\,/\,cred}$ *fly(tweety)* $KB_2 \mid\!\sim_{scept\,/\,cred} \neg$ *fly(tweety)*

Default Logic Approaches to Non-monotonic Reasoning

KB

W = $a \;,\; a \rightarrow b \;,\; c$

Defaults =

1. $\top \Longrightarrow \neg c$
2. $b \Longrightarrow d$
3. $d \Longrightarrow \neg a$
4. $a \Longrightarrow \neg d$

Ordering = 1 > 2 > 3 > 4

1 cannot be applied

2 can be applied = Cn($W \cup \{d\}$)

3 cannot be applied

4 cannot be applied

Week 2 Learning Outcomes

By the end of this week, you will be able to:

- LO1: Evaluate the non-monotonic consequences for three types of non-monotonic logic
- LO2: Construct an argument framework (AF) of arguments and their relations from a knowledge base (KB) of formulas and the inference rules of a logic
- LO3: Apply preferences to determine whether one argument is a legitimate counter-argument to another
- LO4: Apply labelling rules to determine the winning sets of arguments in an AF
- LO5: Relate the claims of winning arguments to KB's non-monotonic logic consequences

Non-monotonic Reasoning as *Argumentation*

- "One can also think of non-monotonic reasoning as a decision procedure for choosing between contradictory **classical** (monotonic) consequences"

- **Generalising this idea**:

"One can also think of non-monotonic reasoning as a decision procedure for choosing between arguments for conflicting claims"

(arguments = premises and rules in support of a claim)

the kinds of decision procedures that (unlike proof theories of non-monotonic logics) are familiar in our everyday uses of argument in discussion/debate …

Non-monotonic Reasoning (Logic Programming) as *Argumentation*

KB_1

1. *penguin(tweety)*
2. *bird(X) :- penguin(X)*
3. *fly(X) :- bird(X), NOT abnormal_fly(X)*

$\mid\!\sim$ *fly(tweety)*

A1 =

1, 2, 3

✓

*fly(tweety) is the claim of the **only** (hence winning or 'justified') argument A1*

Non-monotonic Reasoning (Logic Programming) as *Argumentation*

KB_2= KB_1 ∪ {4}

1. *penguin(tweety)*
2. *bird(X) :- penguin(X)*
3. *fly(X) :- bird(X), NOT abnormal_fly(X)*
4. *abnormal_fly(X) :- penguin(X), NOT DC_comic(X)*

$\not\vdash$ *fly(tweety)*

A2 =

1, 4

A1 =

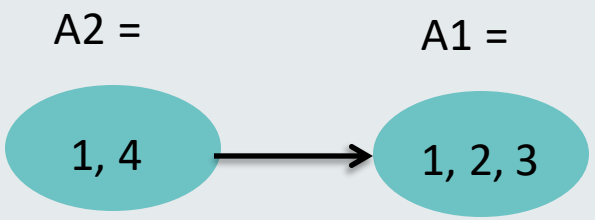
1, 2, 3

Non-monotonic Reasoning (Logic Programming) as *Argumentation*

KB_2= KB_1 ∪ {4}

1. *penguin(tweety)*
2. *bird(X) :- penguin(X)*
3. *fly(X) :- bird(X), NOT abnormal_fly(X)*
4. *abnormal_fly(X) :- penguin(X), NOT DC_comic(X)*

$\not\vdash$ *fly(tweety)*



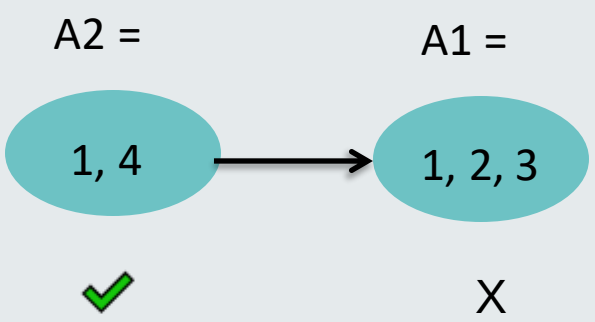
A2 "attacks" (is a counter-argument to) A1 since A2's claim (*abnormal_fly(tweety)*) challenges A1's assumption (NOT *abnormal_fly(tweety)*) that one cannot prove (we do not have an argument for the claim) that Tweety is abnormal with respect to the property of flying

Non-monotonic Reasoning (Logic Programming) as *Argumentation*

KB_2= KB_1 ∪ {4}

1. *penguin(tweety)*
2. *bird(X) :- penguin(X)*
3. *fly(X) :- bird(X), NOT abnormal_fly(X)*
4. *abnormal_fly(X) :- penguin(X), NOT DC_comic(X)*

$\not\vdash$ *fly(tweety)*



A2 is not attacked and so is winning. A1 is attacked by a winning argument and so is losing

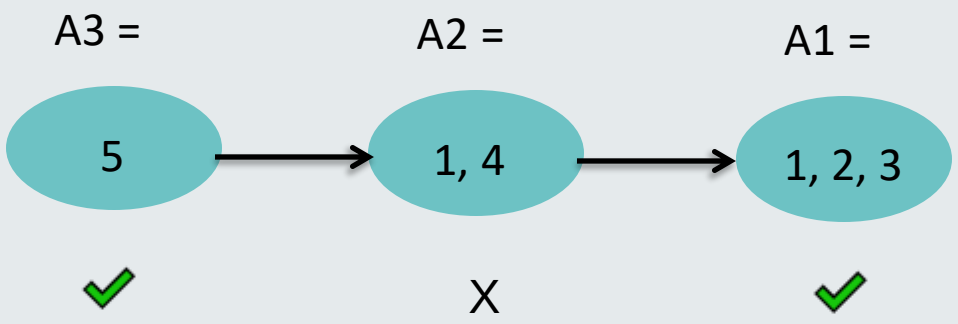
fly(tweety) is not the claim of a winning argument

Non-monotonic Reasoning (Logic Programming) as *Argumentation*

KB_3= KB_2 ∪ {5}

1. *penguin(tweety)*
2. *bird(X) :- penguin(X)*
3. *fly(X) :- bird(X), NOT abnormal_fly(X)*
4. *abnormal_fly(X) :- penguin(X), NOT DC_comic(X)*
5. *DC_comic(tweety)*

$\mid\!\sim$ *fly(tweety)*



A3 is not attacked and so is winning.
A2 is attacked by a winning argument and so is losing
All of A1's attackers (i.e., A2) are losing and so A1 is winning

fly(tweety) ***is*** *the claim of justified argument A1*

Week 2 Learning Outcomes

By the end of this week, you will be able to:

- LO1: Evaluate the non-monotonic consequences for three types of non-monotonic logic

- LO2: Construct an argument framework (AF) of arguments and their relations from a knowledge base (KB) of formulas and the inference rules of a logic

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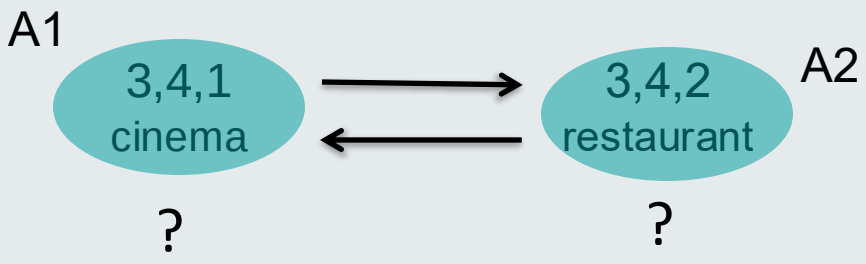
Non-monotonic Reasoning (Logic Programming) as *Argumentation*

1. Bel *birthday* ∧ ***Des*** *celebrate_birthday* ∧ ***Bel*** *NOT closed_cinema* ⟶ ***Goal***(*cinema*)

2. Bel *birthday* ∧ ***Des*** *celebrate_birthday* ∧ ***Bel*** *NOT closed_restaurant* ⟶ ***Goal***(*restaurant*)

3. Bel *birthday*

4. Des *celebrate_birthday*



Neither argument wins out – we are in a genuine dilemma, we are undecided

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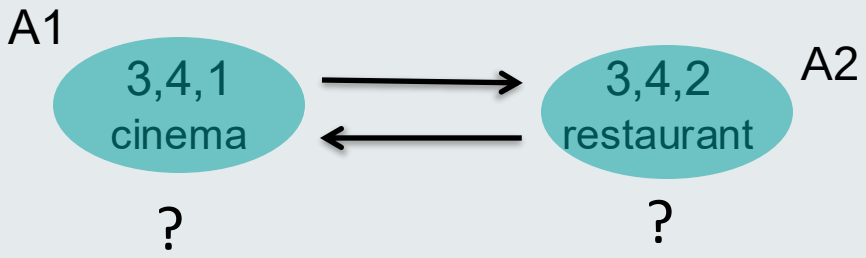

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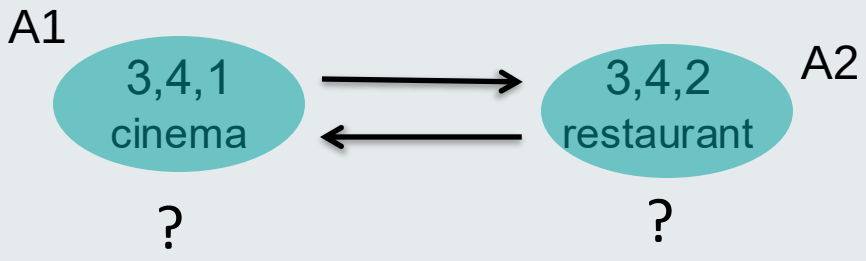

Given 5, A2 is stronger than (preferred to) A1 and so A1 cannot be moved as a counter-argument to A2. Hence attack from A1 to A2 is cancelled out, and only A2 attacks A1 and so A2 wins

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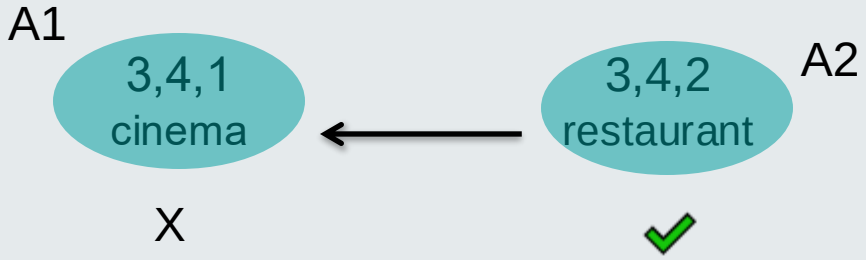

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Non-monotonic Reasoning (Classical Logic) as *Argumentation*

KB

S_1 $\quad a \longrightarrow b \quad a$

S_2 $\quad \neg b \quad c \quad \neg c$

S_3 $\quad d$

Arguments constructed from a stratified classical KB are tuples $[\Delta, \alpha]$ such that

1. $\Delta \subseteq KB, \Delta \vdash \alpha$
2. No proper subset of Δ classically entails α
3. Δ is consistent

A1 = $[\{a \longrightarrow b, \neg b\}, \neg a]$ A2 = $[\{a\}, a]$ A3 = $[\{c\}, c]$ A4 = $[\{\neg c\}, \neg c]$

Non-monotonic Reasoning (Classical Logic) as *Argumentation*

KB

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S_2	$\neg b \quad c \quad \neg c$
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Arguments constructed from a stratified classical KB are tuples $[\Delta, \alpha]$ such that

A = $[\Delta 1, \alpha]$ attacks B = $[\Delta 2, \beta]$ if the claim α of A negates a premise in $\Delta 2$

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Non-monotonic Reasoning (Classical Logic) as *Argumentation*

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An attack succeeds if the attacked premise (e.g., a in A2) is not ordered above any of the premises in the attacking argument ($a \longrightarrow b, \neg b$)

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A5 = $[\{a, a \longrightarrow b\}, b]$ ✓

A1 = $[\{a \longrightarrow b, \neg b\}, \neg a]$ X

A2 = $[\{a\}, a]$ ✓

A3 = $[\{c\}, c]$?

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Non-monotonic Reasoning (Classical Logic) as *Argumentation*

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S_2	$\neg b \quad c \quad \neg c$
S_3	d

$KB_2 \mid\!\sim_{cred} c\ ,\ \neg c$

$KB_2 \mid\!\sim_{scept\,/\,cred} a\ ,\ b$

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Non-monotonic Reasoning as *Argumentation*

- We have seen how arguments can be constructed from a KB and related to each by a binary attack relation whose success may depend on preferences amongst arguments

- The arguments constructed from a KB can then be organised into a directed graph in which the arguments are nodes and the successful attacks are directed binary relations amongst nodes

- We thus obtain a Dung Argumentation Framework [1] $< Args, Attacks >$ where $Attacks = \{ (X,Y) \mid X,Y \in Args \}$

$A5 = [\{a, a \longrightarrow b\}, b]$ ✔ $A1 = [\{a \longrightarrow b, \neg b\}, \neg a]$ X $A2 = [\{a\}, a]$ ✔ $A3 = [\{c\}, c]$? $A4 = [\{\neg c\}, \neg c]$?

1. P.M. Dung, On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n–person games, Artificial Intelligence, 77 (1995) 321–357

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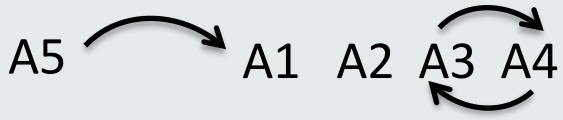
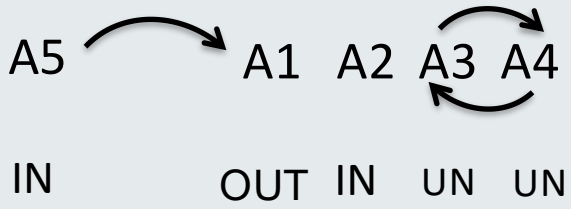


Labelling Argumentation Frameworks

- Given an AF < *Args,Attacks* > of arguments related by attacks, what are the rules for assigning winning losing and undecided 'labels' to arguments ?



A labelling $\mathscr{L}$ results from applying a *total function LAB*(AF) *that assigns to each X* $\in$ *Args, either IN (winning), OUT (losing) or UN (undecided), and such that:*

1. *X is IN iff* $\forall$ *(Y,X)* $\in$ *Attacks* : *Y is OUT*
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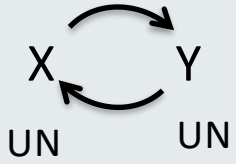
Labelling Argumentation Frameworks

- *LAB* may return many labellings (*LAB* is a one-to-many function)

$\mathscr{L}_1$ =



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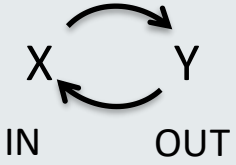

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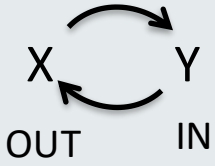

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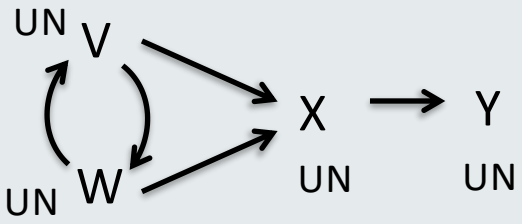

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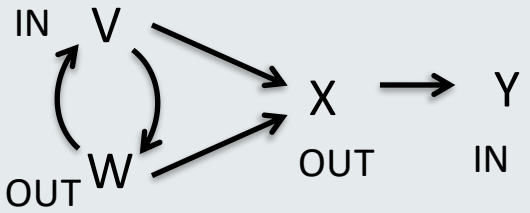

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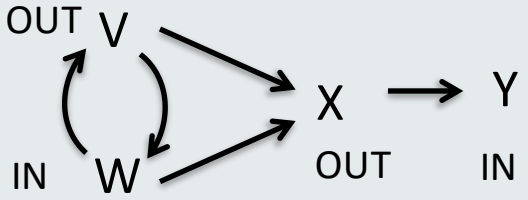

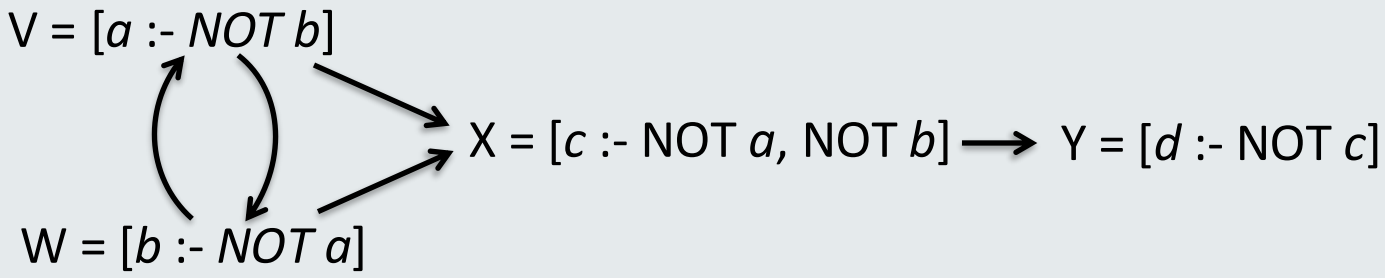

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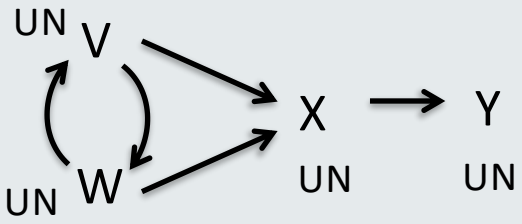

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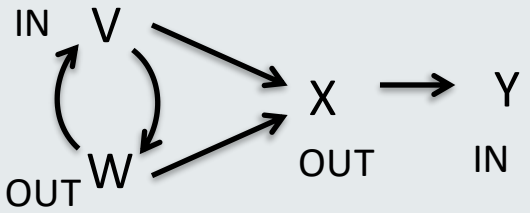

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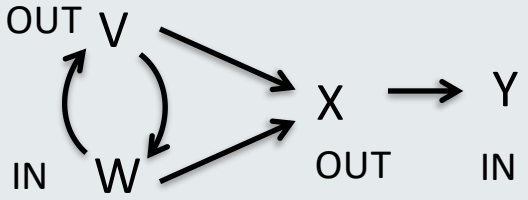

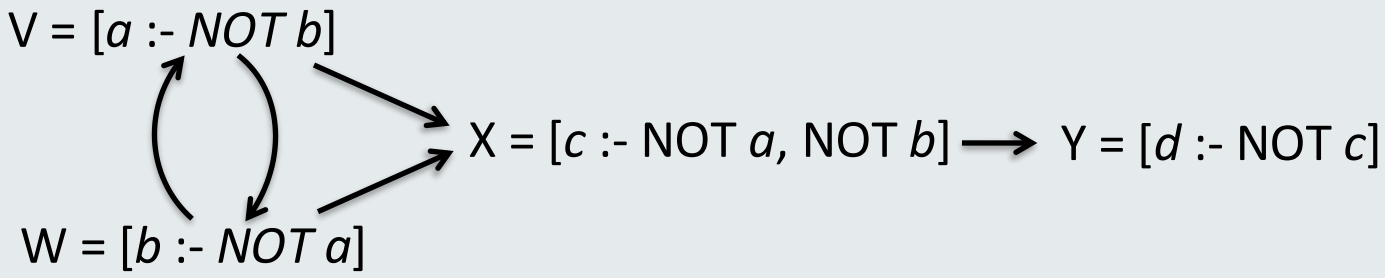

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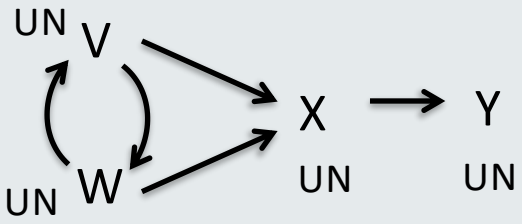

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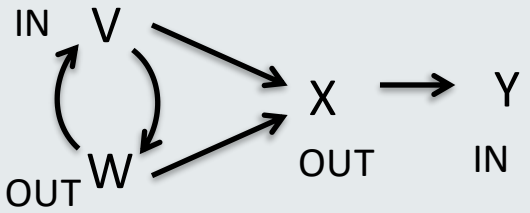

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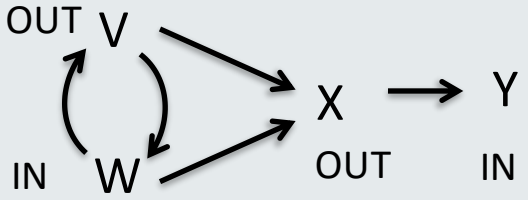

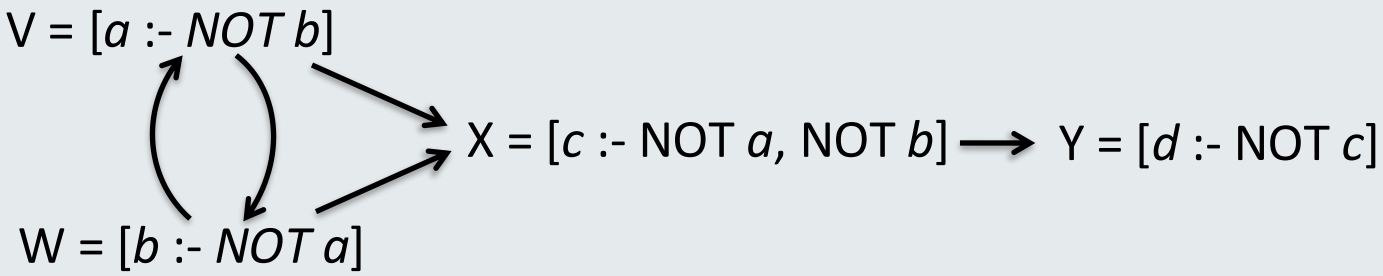

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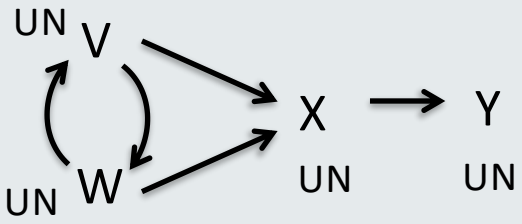

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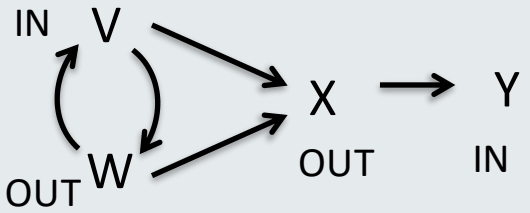

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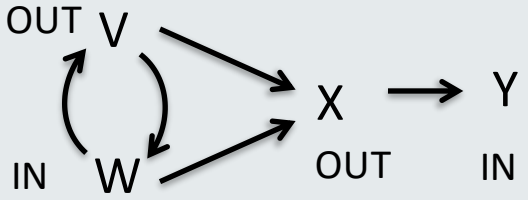

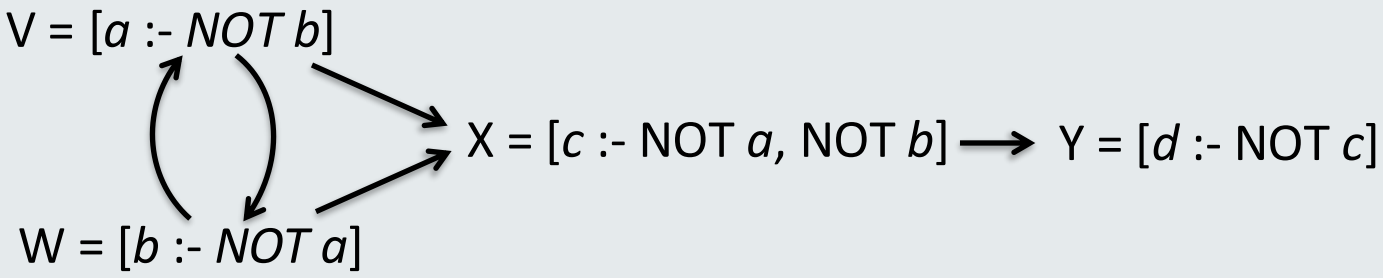

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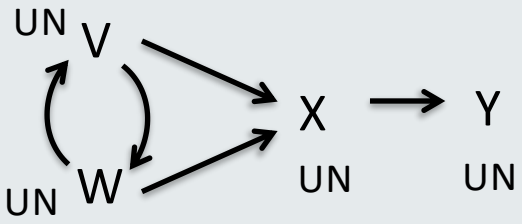

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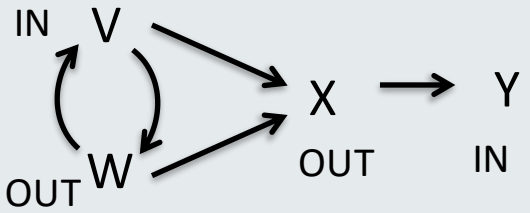

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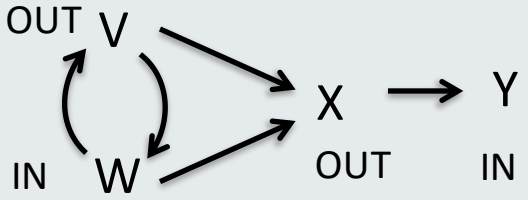

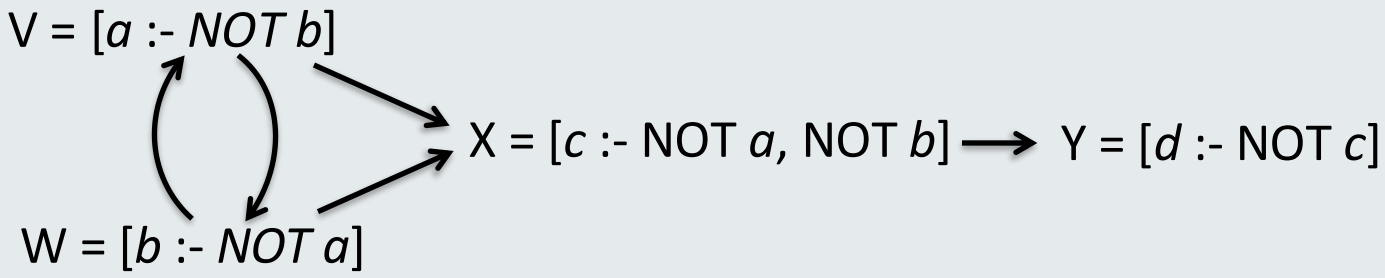

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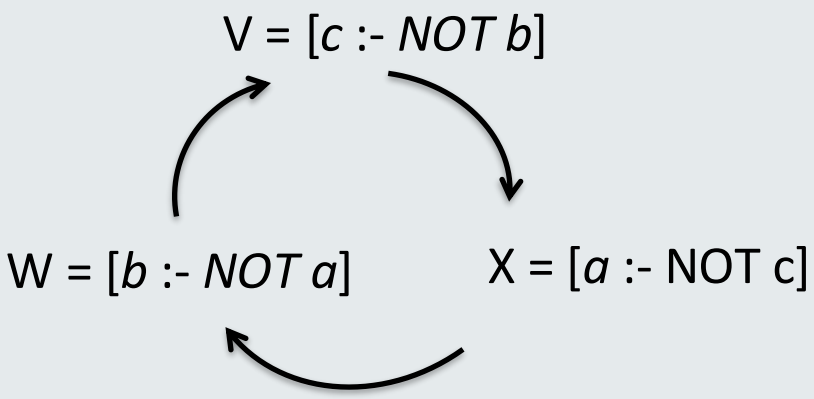
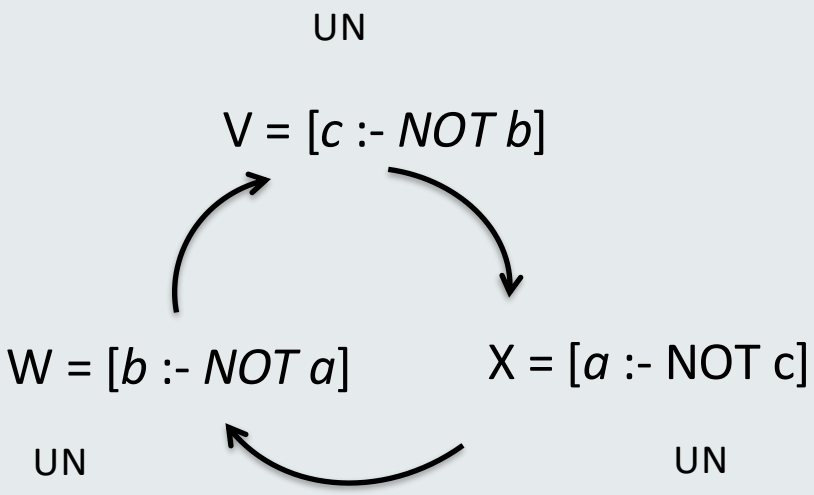
Labelling Argumentation Frameworks (More Examples)

- What are the preferred/grounded labellings for



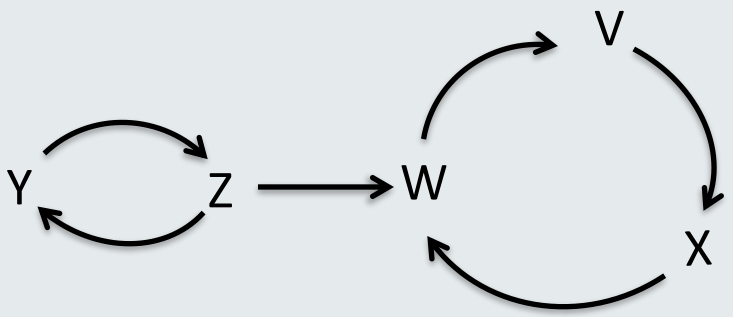
Labelling Argumentation Frameworks (More Examples)

- Only one preferred/grounded labelling:



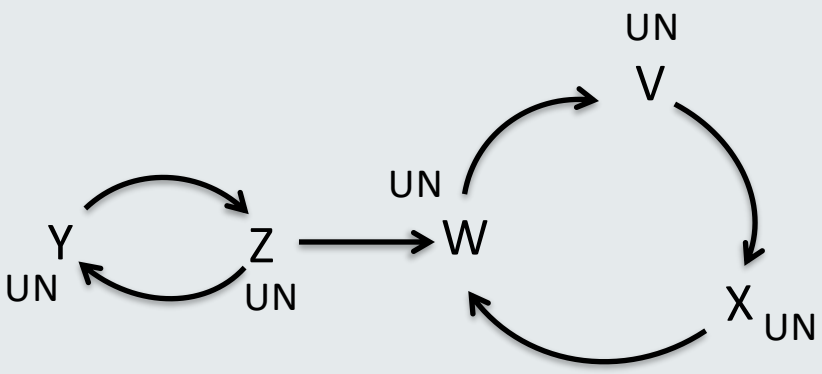
Labelling Argumentation Frameworks (More Examples)

- What are the preferred/grounded labellings for:



Labelling Argumentation Frameworks (More Examples)

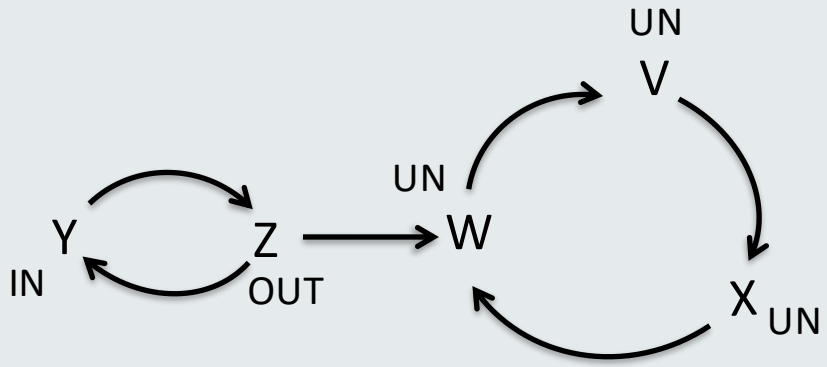
- The grounded labelling is:



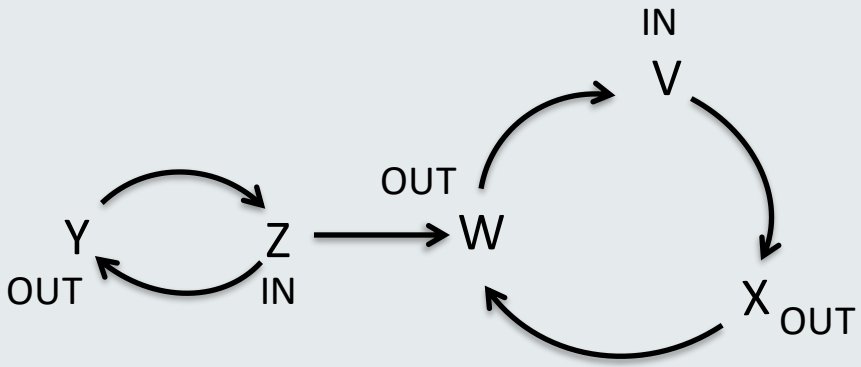
Labelling Argumentation Frameworks (More Examples)

- There are two preferred labellings:

$\mathscr{L}_1$ =



$\mathscr{L}_2$ =



- Notice that $\mathscr{L}_1$ is a preferred labelling even though it has less IN arguments then $\mathscr{L}_2$

Week 2 Summary

By the end of this week, you will be able to:

- LO1: Evaluate the non-monotonic consequences for three types of non-monotonic logic

- LO2: Construct an argument framework (AF) of arguments and their relations from a knowledge base (KB) of formulas and the inference rules of a logic

- LO3: Apply preferences to determine whether one argument is a legitimate counter-argument to another

- LO4: Apply labelling rules to determine the winning sets of arguments in an AF

- LO5: Relate the claims of winning arguments to KB's non-monotonic logic consequences

Next week

- Week 3

  - *Argument Game Proof theories* to decide whether an argument is winning

  - From argument game proof theories to *dialogues for* ***distributed*** *non-monotonic reasoning*: Multiple agents communicating so as to pool their knowledge and reason together

  - The importance of dialogue in the dawning age of Artificial Intelligence